ÁLGEBRA LINEAL

MATRIZ ADJUNTA

DEFINICIÓN

Sea $A=(a_{ij})\in M_n(\mathbb{R})$ Se define la matriz adjunta de A como:

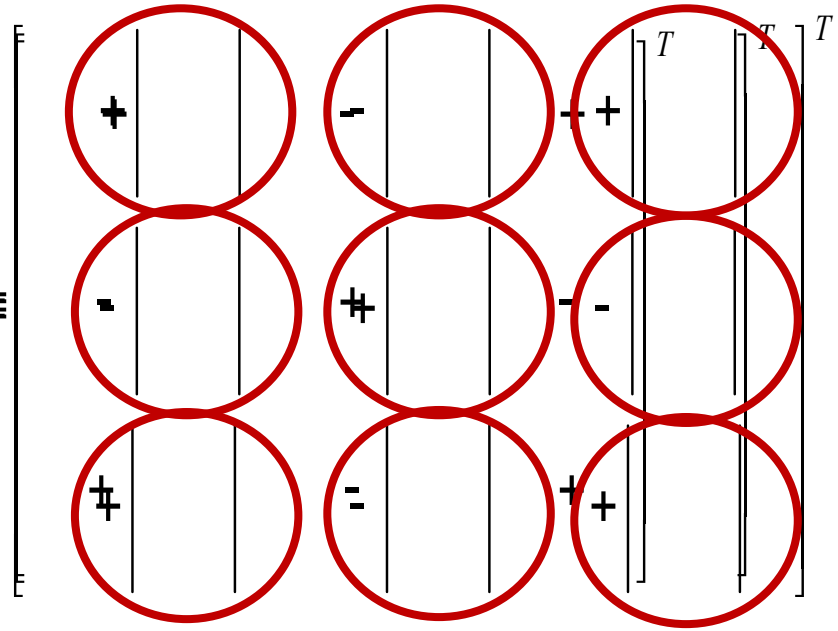
$$adj(A)=\begin{bmatrix} A_{11} & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} & \dots & A_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ A_{n1} & A_{n2} & \dots & A_{nn} \end{bmatrix}^T$$

Donde A_{ij} es el cofactor de a_{ij}

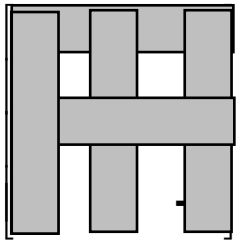
EJEMPLO

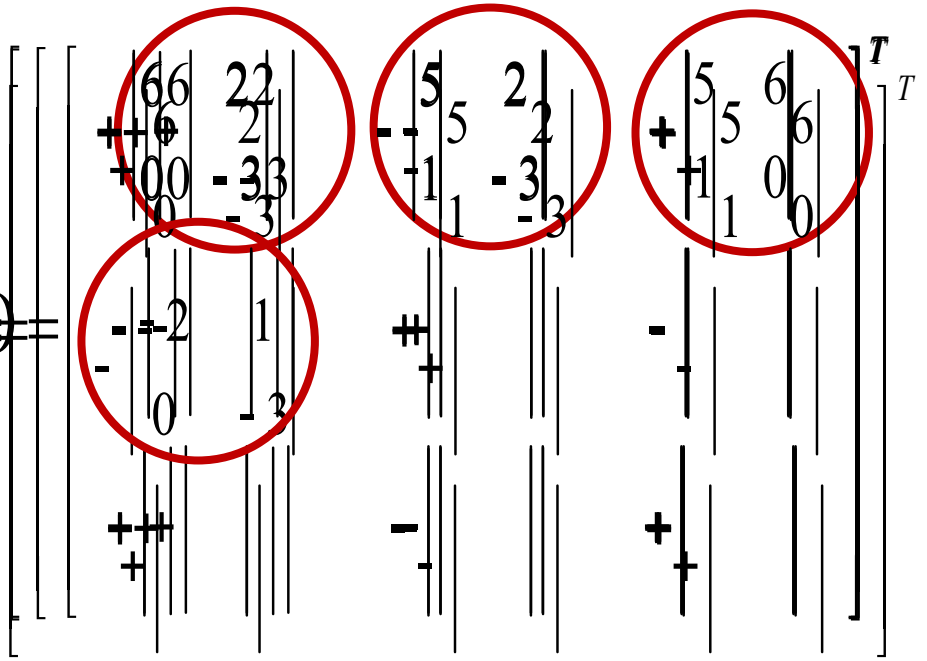
Sea $A = \begin{bmatrix} 3 & -2 & 1 \\ 5 & 6 & 2 \\ 1 & 0 & -3 \end{bmatrix}$ Determine la matriz adjunta de A

$$adj(A) \equiv$$



EJEMPLO

Sea $A =$ 

$$adj(A) =$$ 

MATRIZ ADJUNTA

Siguiendo el mismo procedimiento....

$$adj(A) = \begin{bmatrix} +\begin{vmatrix} 6 & 2 \\ 0 & -3 \end{vmatrix} & -\begin{vmatrix} 5 & 2 \\ 1 & -3 \end{vmatrix} & +\begin{vmatrix} 5 & 6 \\ 1 & 0 \end{vmatrix} \\ -\begin{vmatrix} -2 & 1 \\ 0 & -3 \end{vmatrix} & +\begin{vmatrix} 3 & 1 \\ 1 & -3 \end{vmatrix} & -\begin{vmatrix} 3 & -2 \\ 1 & 0 \end{vmatrix} \\ +\begin{vmatrix} -2 & 1 \\ 6 & 2 \end{vmatrix} & -\begin{vmatrix} 3 & 1 \\ 5 & 2 \end{vmatrix} & +\begin{vmatrix} 3 & -2 \\ 5 & 6 \end{vmatrix} \end{bmatrix}^T = \begin{bmatrix} -18 & 17 & -6 \\ -6 & -10 & -2 \\ -10 & -1 & 28 \end{bmatrix}^T = \begin{bmatrix} -18 & -6 & -10 \\ 17 & -10 & -1 \\ -6 & -2 & 28 \end{bmatrix}$$